

MySQL

```
#!/bin/bash
# =====
# ■■■3■■■MySQL■■■■■■■■■■
# ■■■■■■MySQL■■■■■■■■■■■■■■■■■■■■
# ■■■■■■■■
# ■■■■20231035109
# =====
# ■■■■■■
RED='\033[0;31m'
GREEN='\033[0;32m'
YELLOW='\033[1;33m'
BLUE='\033[0;34m'
CYAN='\033[0;36m'
NC='\033[0m'
# MySQL■■■■■
MYSQL_ROOT_PASSWORD="Root@123456"
MYSQL_VERSION="8.0"
MYSQL_INSTALL_LOG="/workspace/linux_course_design/mysql_install.log"
MYSQL_DATA_DIR="/var/lib/mysql"
# ■■■■■■■■
init_mysql_log() {
echo "===== " > "$MYSQL_INSTALL_LOG"
echo "MySQL■■■■■" >> "$MYSQL_INSTALL_LOG"
echo "■■■■■: $(date '+%Y-%m-%d %H:%M:%S')" >> "$MYSQL_INSTALL_LOG"
echo "===== " >> "$MYSQL_INSTALL_LOG"
}
# ■■■■■■
mysql_log() {
local msg="$1"
local timestamp=$(date '+%Y-%m-%d %H:%M:%S')
echo "[$timestamp] $msg" | tee -a "$MYSQL_INSTALL_LOG"
}
# =====
# ■■■1■■■■■■■■■■
# ■■■■■■
# 1. ■■■■■■■■■CentOS/Ubuntu/■■■■
# 2. ■■■■■■■■MySQL■■■■■■■■■■
# 3. ■■■■■■■■■■■■■■■■■■■■■
# 4. ■■■■■■■■■
# =====
check_environment() {
echo -e "${CYAN}===== ${NC}"
echo -e "${CYAN} ■■■■■■${NC}"
echo -e "${CYAN}===== ${NC}"
mysql_log "[■■■1] ■■■■■■■■■..."
# 1.1 ■■■■■■■■■
echo -e "\n${YELLOW}[■■■] ■■■■■■■■■...${NC}"
if [ -f /etc/os-release ]; then
. /etc/os-release
OS_NAME=$NAME
OS_VERSION=$VERSION_ID
echo -e "${GREEN} ■■■■■: $OS_NAME $OS_VERSION${NC}"
mysql_log "■■■■■: $OS_NAME $OS_VERSION"
else
echo -e "${RED} [■■■] ■■■■■■■■■${NC}"
return 1
fi
# 1.2 ■■■■■■■■MySQL
echo -e "${YELLOW}[■■■] MySQL■■■■■...${NC}"
if command -v mysql &>/dev/null; then
MYSQL_INSTALLED_VERSION=$(mysql --version)
echo -e "${YELLOW} MySQL■■■■■: $MYSQL_INSTALLED_VERSION${NC}"
mysql_log "MySQL■■■■■: $MYSQL_INSTALLED_VERSION"
read -p "■■■■■■■■■■MySQL(y/n): " reinstall
if [ "$reinstall" != "y" ] && [ "$reinstall" != "Y" ]; then
echo -e "${YELLOW}■■■■■■■■■■...${NC}"
return 0
fi
else
echo -e "${GREEN} MySQL■■■■■■■■■■■${NC}"
mysql_log "MySQL■■■■■■■■■■■"
```

```

fi
# 1.3 MySQL512MB
echo -e "${YELLOW}[██] .....${NC}"
total_mem=$(free -m | awk '/Mem:/ {print $2}')
echo -e "${GREEN}██: ${total_mem}MB${NC}"
if [ "$total_mem" -lt 512 ]; then
echo -e "${RED} [██] 512MBMySQL${NC}"
mysql_log "██: 512MB ($total_mem)"
fi
mysql_log "██: ${total_mem}MB"
# 1.4 MySQL500MB
echo -e "${YELLOW}[██] .....${NC}"
available_space=$(df -BG / | awk 'NR==2 {print $4}' | tr -d 'G')
echo -e "${GREEN}██: ${available_space}GB${NC}"
if [ "$available_space" -lt 1 ]; then
echo -e "${RED} [██] .....${NC}"
return 1
fi
mysql_log "██: ${available_space}GB"
# 1.5
echo -e "${YELLOW}[██] .....${NC}"
if ping -c 1 -W 3 mirrors.aliyun.com &>/dev/null; then
echo -e "${GREEN}██: .....${NC}"
mysql_log "██: ██"
else
echo -e "${RED} [██] .....${NC}"
mysql_log "██: ....."
fi
echo -e "\n${GREEN}[██] .....${NC}"
return 0
}
# =====
# 2MySQL
#
# 1.
# 2. MySQLYUM/APT
# 3. MySQL Server
# 4.
# =====
install_mysql() {
echo -e "${CYAN}===== ${NC}"
echo -e "${CYAN} MySQL${NC}"
echo -e "${CYAN}===== ${NC}"
mysql_log "[2] MySQL..."
#
case "$OS_NAME" in
*"Ubuntu"*|"Debian"*)
install_mysql_ubuntu
;;
*"CentOS"*|"Red Hat"*|"Rocky"*|"AlmaLinux"*)
install_mysql_centos
;;
*)
echo -e "${RED}[██] : $OS_NAME${NC}"
echo -e "${YELLOW}MySQL${NC}"
return 1
;;
esac
}
# Ubuntu/DebianMySQL
install_mysql_ubuntu() {
echo -e "${YELLOW}[Ubuntu/Debian] .....${NC}"
# apt
sudo apt-get update -y
echo -e "${YELLOW}[MySQL Server...]${NC}"
mysql_log "aptMySQL..."
# DEBIAN_FRONTEND=noninteractive
# root
sudo DEBIAN_FRONTEND=noninteractive apt-get install -y \
mysql-server mysql-client
if [ $? -eq 0 ]; then
echo -e "${GREEN}[██] MySQL${NC}"
mysql_log "MySQL (Ubuntu/Debian)"
else

```

```
echo -e "${RED}[██] MySQL${NC}"
mysql_log "MySQL███"
return 1
fi
}
# CentOS/RHEL██MySQL██
install_mysql_centos() {
echo -e "${YELLOW}[CentOS/RHEL] ██MySQL YUM...${NC}"
# █████MySQL████
# █CentOS 7+████████MySQL
if [ ! -f /etc/yum.repos.d/mysql-community.repo ]; then
# █MySQL████MySQL 8.0██
sudo yum install -y https://dev.mysql.com/get/mysql80-community-release-el7-11.noarch.rpm 2>/dev/null || \
{
echo -e "${YELLOW}████████████████MariaDB...${NC}"
# █████MySQL████████████████MariaDB██
sudo yum install -y mariadb-server mariadb
if [ $? -eq 0 ]; then
echo -e "${GREEN}[██] MariaDBMySQL████████${NC}"
mysql_log "MariaDB████ (CentOS)"
return 0
fi
}
fi
echo -e "${YELLOW}[████MySQL Server...]${NC}"
mysql_log "██yum██MySQL..."
# █MySQL Server
sudo yum install -y mysql-server mysql
if [ $? -eq 0 ]; then
echo -e "${GREEN}[██] MySQL██████${NC}"
mysql_log "MySQL████ (CentOS)"
else
echo -e "${RED}[██] MySQL██████${NC}"
mysql_log "MySQL████"
return 1
fi
}
# =====
# █3██MySQL████
# █████
# 1. █MySQL█bin████PATH████
# 2. ███etc/profile.d/mysql.sh████████
# 3. █MySQL████████UTF-8
# 4. █MySQL████████
# =====
configure_mysql_env() {
echo -e "${CYAN}=-----${NC}"
echo -e "${CYAN} █MySQL██${NC}"
echo -e "${CYAN}=-----${NC}"
mysql_log "[██3] █MySQL████..."
# 3.1 █MySQL████
if command -v mysql &>/dev/null; then
MYSQL_BIN_DIR=$(dirname $(which mysql))
echo -e "${GREEN}MySQL██: $MYSQL_BIN_DIR${NC}"
else
MYSQL_BIN_DIR="/usr/bin"
echo -e "${YELLOW}████MySQL██: $MYSQL_BIN_DIR${NC}"
fi
# 3.2 ██████████
# ███etc/profile.d/████████████████
ENV_FILE="/etc/profile.d/mysql.sh"
echo -e "${YELLOW}[██] ███████...${NC}"
sudo tee "$ENV_FILE" > /dev/null << EOF
#!/bin/bash
# MySQL██████
# █████: $(date '+%Y-%m-%d %H:%M:%S')
# █MySQL██PATH
export PATH=$PATH:$MYSQL_BIN_DIR:$MYSQL_BIN_DIR/../sbin
# MySQL██████
export MYSQL_HOME=${MYSQL_BIN_DIR%/bin}
export DATADIR=$MYSQL_DATA_DIR
# MySQL██████
export MYSQL_PAGER=less
```

```
export MYSQL_EDITOR=vim
EOF
# ■■■■■■■■
sudo chmod +x "$ENV_FILE"
# 3.3 ■■■■■■■■
source "$ENV_FILE"
export PATH=$PATH:$MYSQL_BIN_DIR
echo -e "${GREEN}[■] ■■■■■■■■${NC}"
echo -e "    ■■■■: $ENV_FILE"
mysql_log "■■■■■■■■: $ENV_FILE"
# 3.4 ■■■MySQL■■■■■■■
echo -e "${YELLOW}[■] ■■■MySQL■■■■...${NC}"
sudo systemctl enable mysqld 2>/dev/null || sudo systemctl enable mariadb 2>/dev/null
echo -e "${GREEN}[■] MySQL■■■■■■■■■${NC}"
mysql_log "MySQL■■■■■■■■■"
}
# =====
# ■■■4■■■■MySQL■■root■■
# ■■■■■■
# 1. ■■■MySQL■■■
# 2. ■■■■■■■■■■mysql_secure_installation■
# 3. ■■■root■■■■■
# 4. ■■■■■■■■■■
# =====
initialize_mysql() {
echo -e "${CYAN}=-----=${NC}"
echo -e "${CYAN}   ■■■MySQL■■■${NC}"
echo -e "${CYAN}=-----=${NC}"
mysql_log "[■■4]   ■■■MySQL..."
# 4.1 ■■■MySQL■■■
echo -e "${YELLOW}[■] MySQL■■■...${NC}"
sudo systemctl start mysqld 2>/dev/null || sudo systemctl start mariadb 2>/dev/null
sleep 3 # ■■■■■■■■
# ■■■■■■
if sudo systemctl is-active --quiet mysqld 2>/dev/null || \
sudo systemctl is-active --quiet mariadb 2>/dev/null; then
echo -e "${GREEN}[■] MySQL■■■■■■■${NC}"
mysql_log "MySQL■■■■■■■■■"
else
echo -e "${RED}[■] MySQL■■■■■■■■■${NC}"
mysql_log "MySQL■■■■■■■■■"
return 1
fi
# 4.2 ■■■■■■■■MySQL 8.0■■■■■■■■■■■■■■■
INITIAL_PASSWORD=""
if [ -f /var/log/mysqld.log ]; then
INITIAL_PASSWORD=$(sudo grep 'temporary password' /var/log/mysqld.log | awk '{print $NF}')
if [ -n "$INITIAL_PASSWORD" ]; then
echo -e "${YELLOW}[■] ■■■■■■■■■■${NC}"
fi
fi
# 4.3 ■■■root■■■
echo -e "${YELLOW}[■] ■■■root■■■■...${NC}"
if [ -n "$INITIAL_PASSWORD" ]; then
# MySQL 8.0■■■■■■■■■■■■■■■
mysql -u root -p"$INITIAL_PASSWORD" --connect-expired-password \
-e "ALTER USER 'root'@'localhost' IDENTIFIED BY '$MYSQL_ROOT_PASSWORD';" 2>/dev/null
else
# MariaDB■■■■■■■■■
mysql -u root -e "ALTER USER 'root'@'localhost' IDENTIFIED BY '$MYSQL_ROOT_PASSWORD';" 2>/dev/null || \
mysqladmin -u root password "$MYSQL_ROOT_PASSWORD" 2>/dev/null
fi
if [ $? -eq 0 ]; then
echo -e "${GREEN}[■] root■■■■■■■■■${NC}"
echo -e "    root■■■: $MYSQL_ROOT_PASSWORD"
mysql_log "root■■■■■■■■■"
else
echo -e "${YELLOW}[■] ■■■■■■■■■■...${NC}"
# ■■■■■■■■■■mysql.user■
sudo mysqld --skip-grant-tables --user=mysql &
sleep 2
mysql -u root -e "UPDATE mysql.user SET authentication_string=PASSWORD('$MYSQL_ROOT_PASSWORD') WHERE
User='root'; FLUSH PRIVILEGES;" 2>/dev/null || \
mysql -u root -e "SET PASSWORD FOR 'root'@'localhost' = PASSWORD('$MYSQL_ROOT_PASSWORD');" 2>/dev/null
```

```
sudo pkill -9 mysqld
sudo systemctl start mysqld 2>/dev/null || sudo systemctl start mariadb 2>/dev/null
mysql_log "■■■■■■■■■■■"
fi
# 4.4 ■■■■■■■■■■
echo -e "${YELLOW}[██] ■■■■■...${NC}"
mysql -u root -p"$MYSQL_ROOT_PASSWORD" 2>/dev/null << EOF
-- ■■■■■■
DELETE FROM mysql.user WHERE User='';
-- ■■root■■■■■■■■■■■
DELETE FROM mysql.user WHERE User='root' AND Host NOT IN ('localhost', '127.0.0.1', '::1');
-- ■■■■■■
DROP DATABASE IF EXISTS test;
DELETE FROM mysql.db WHERE Db='test' OR Db='test\\_%';
-- ■■■■
FLUSH PRIVILEGES;
EOF
echo -e "${GREEN}[██] MySQL■■■■■■■${NC}"
mysql_log "MySQL■■■■■■■"
}
# =====
# ■■5■■■MySQL■■
# ■■■■■
# 1. ■■■■■root■■■■■■MySQL
# 2. ■■MySQL■■■■■■■■■■■
# 3. ■■■■■■■■■■
# =====
login_mysql_test() {
echo -e "${CYAN}=-----${NC}"
echo -e "${CYAN} ■■MySQL■■■${NC}"
echo -e "${CYAN}=-----${NC}"
mysql_log "[■■5] ■■MySQL■■■..."
echo -e "${YELLOW}[██] ■■MySQL■■■...${NC}"
# ■■■■■MySQL
if mysql -u root -p"$MYSQL_ROOT_PASSWORD" -e "SELECT VERSION();" 2>/dev/null; then
echo -e "\n${GREEN}[██] MySQL■■■■■■■${NC}"
mysql_log "MySQL■■■■■■■"
# ■■MySQL■■■■■
echo -e "\n${YELLOW}--- MySQL■■■■ ---${NC}"
mysql -u root -p"$MYSQL_ROOT_PASSWORD" -e "SELECT VERSION() AS 'MySQL■■■';" 2>/dev/null
# ■■■■■
echo -e "\n${YELLOW}--- ■■■■ ---${NC}"
mysql -u root -p"$MYSQL_ROOT_PASSWORD" -e "SELECT USER() AS '■■■■';" 2>/dev/null
# ■■■■■
echo -e "\n${YELLOW}--- ■■■■ ---${NC}"
sudo systemctl status mysqld 2>/dev/null | head -5 || sudo systemctl status mariadb 2>/dev/null | head
-5
return 0
else
echo -e "${RED}[██] MySQL■■■■■■■■■■■■■■■■■■■■■${NC}"
mysql_log "MySQL■■■■■■■"
return 1
fi
}
# =====
# ■■6■■■■■■■■■■■
# ■■■■■
# 1. ■■■■■■■■■■■■■■■■■■■■■■
# 2. ■■■■■■■■■■■■■■■■■■■■■■
# 3. ■■■■■■■■■■
# 4. ■■■■■■■■■■
# =====
create_database_and_tables() {
echo -e "${CYAN}=-----${NC}"
echo -e "${CYAN} ■■■■■■${NC}"
echo -e "${CYAN}=-----${NC}"
mysql_log "[■■6] ■■■■■■■■■■..."
# ■■■■■
DB_NAME="student_management"
echo -e "${YELLOW}[██] ■■■: $DB_NAME${NC}"
# ■■SQL■■■■■■■■■■■
mysql -u root -p"$MYSQL_ROOT_PASSWORD" 2>/dev/null << EOSQL
-- =====
-- ■■■■■■■■■■
```

```
-- #####: $(date '+%Y-%m-%d %H:%M:%S')
-- =====
-- #####
DROP DATABASE IF EXISTS $DB_NAME;
-- #####UTF-8#####
CREATE DATABASE $DB_NAME
DEFAULT CHARACTER SET utf8mb4
COLLATE utf8mb4_unicode_ci;
-- #####
USE $DB_NAME;
-- -----
-- ■1: students#####
-- #####:
-- student_id: #####
-- name: ■■
-- gender: ■■
-- age: ■■
-- class_name: ■■
-- phone: #####
-- create_time: #####
-- -----
CREATE TABLE students (
student_id INT AUTO_INCREMENT PRIMARY KEY COMMENT '■■',
name VARCHAR(50) NOT NULL COMMENT '■■',
gender ENUM('■', '■') DEFAULT '■' COMMENT '■■',
age INT CHECK (age BETWEEN 15 AND 30) COMMENT '■■',
class_name VARCHAR(50) COMMENT '■■',
phone VARCHAR(20) COMMENT '■■■■',
create_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP COMMENT '■■■■'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COMMENT='■■■■';
-- -----
-- ■2: courses#####
-- #####:
-- course_id: #####
-- course_name: ■■■■
-- credit: ■■
-- teacher: ■■■■
-- -----
CREATE TABLE courses (
course_id INT AUTO_INCREMENT PRIMARY KEY COMMENT '■■■■',
course_name VARCHAR(100) NOT NULL COMMENT '■■■■',
credit DECIMAL(3,1) NOT NULL DEFAULT 1.0 COMMENT '■■',
teacher VARCHAR(50) NOT NULL COMMENT '■■■■'
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COMMENT='■■■■';
-- -----
-- ■3: scores#####
-- #####:
-- score_id: #####ID#####
-- student_id: #####students■■
-- course_id: #####courses■■
-- score: ■■■0-100■
-- exam_time: ■■■■
-- #####
-- -----
CREATE TABLE scores (
score_id INT AUTO_INCREMENT PRIMARY KEY COMMENT '■■ID',
student_id INT NOT NULL COMMENT '■■',
course_id INT NOT NULL COMMENT '■■■■',
score DECIMAL(5,2) CHECK (score BETWEEN 0 AND 100) COMMENT '■■',
exam_time DATE COMMENT '■■■■',
-- #####
CONSTRAINT fk_student FOREIGN KEY (student_id)
REFERENCES students(student_id) ON DELETE CASCADE,
CONSTRAINT fk_course FOREIGN KEY (course_id)
REFERENCES courses(course_id) ON DELETE CASCADE
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COMMENT='■■■■';
EOSQL
if [ $? -eq 0 ]; then
echo -e "${GREEN}[■■] ■■■ '$DB_NAME' #####${NC}"
mysql_log "■■■ $DB_NAME ■■■■"
else
echo -e "${RED}[■■] #####${NC}"
mysql_log "■■■■■■■■"
return 1
fi
```

```

fi
}
# =====
# ■■■7■■■■■■■■
# ■■■■■■
# 1. ■■■■■■■■■■
# 2. ■■■■■■■■■■
# 3. ■■■■■■■■■■
# 4. ■■■■■■■■■■
# =====
insert_sample_data() {
echo -e "${CYAN}===== ${NC}"
echo -e "${CYAN} ■■■■■■ ${NC}"
echo -e "${CYAN}===== ${NC}"
mysql_log "[■■■7] ■■■■■■..."
DB_NAME="student_management"
echo -e "${YELLOW}[■■■] ■■■■■■... ${NC}"
# ■■■■■■ SQL
mysql -u root -p"${MYSQL_ROOT_PASSWORD}" "$DB_NAME" 2>/dev/null << EOSQL
-- =====
-- ■■■■■■
-- =====
USE $DB_NAME;
-- ■■■■■■
INSERT INTO students (name, gender, age, class_name, phone) VALUES
('■■■', '■', 20, '■■■■2301', '13800138001'),
('■■■', '■', 19, '■■■■2301', '13800138002'),
('■■■', '■', 21, '■■■■2302', '13800138003'),
('■■■', '■', 20, '■■■■2302', '13800138004'),
('■■■', '■', 19, '■■■■2301', '13800138005'),
('■■■', '■', 22, '■■■■2301', '13800138006');
-- ■■■■■■
INSERT INTO courses (course_name, credit, teacher) VALUES
('Linux■■■■', 3.5, '■■■■'),
('■■■■■■', 4.0, '■■■■'),
('■■■■■■', 3.0, '■■■■'),
('Java■■■■', 4.0, '■■■■'),
('■■■■■■■■', 3.5, '■■■■');
-- ■■■■■■
INSERT INTO scores (student_id, course_id, score, exam_time) VALUES
(1, 1, 92.50, '2024-01-15'),
(1, 2, 88.00, '2024-01-16'),
(1, 3, 76.50, '2024-01-17'),
(2, 1, 85.00, '2024-01-15'),
(2, 2, 91.50, '2024-01-16'),
(2, 4, 78.00, '2024-01-18'),
(3, 1, 95.00, '2024-01-15'),
(3, 3, 82.50, '2024-01-17'),
(3, 5, 89.00, '2024-01-19'),
(4, 2, 93.00, '2024-01-16'),
(4, 4, 86.50, '2024-01-18'),
(5, 1, 79.00, '2024-01-15'),
(5, 5, 94.50, '2024-01-19'),
(6, 2, 87.00, '2024-01-16'),
(6, 3, 90.00, '2024-01-17');
EOSQL
if [ $? -eq 0 ]; then
echo -e "${GREEN}[■■■] ■■■■■■ ${NC}"
mysql_log "■■■■■■■■"
# ■■■■■■
echo -e "\n${YELLOW}--- ■■■■ --- ${NC}"
mysql -u root -p"${MYSQL_ROOT_PASSWORD}" "$DB_NAME" -e "
SELECT 'students' AS '■■■', COUNT(*) AS '■■■■' FROM students
UNION ALL
SELECT 'courses', COUNT(*) FROM courses
UNION ALL
SELECT 'scores', COUNT(*) FROM scores;
" 2>/dev/null
# ■■■■■■
echo -e "\n${YELLOW}--- ■■■■■■ --- ${NC}"
mysql -u root -p"${MYSQL_ROOT_PASSWORD}" "$DB_NAME" -e "SELECT * FROM students LIMIT 5;" 2>/dev/null
else
echo -e "${RED}[■■■] ■■■■■■ ${NC}"
mysql_log "■■■■■■■■"

```

[illegible]

main_menu